

# Storage Architecture Framework

## Redeployment Proposal Draft

- Analysis of current operating storage framework
- Alternate redesign deployment advised

Authored by:

N. Kim Tran  
AI / HPC Infrastructure Systems Engineer

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# Mock re-creation of existing storage environment

```
Personalities : [raid4] [raid5] [raid6]
md0 : active raid5 zd112[8] zd96[6] zd80[5] zd64[4] zd48[3] zd32[2] zd16[1] zd0[0]
      1060864 blocks super 1.2 level 5, 512k chunk, algorithm 2 [8/8] [UUUUUUUU]
      bitmap: 0/1 pages [0KB], 65536KB chunk

unused devices: <none>
```

Mdadm : MD0  
Raid 5

One disk failure  
tolerance means  
data in critical  
state before data  
loss if another  
fails.

```
NAME      MAJ:MIN RM  SIZE RO TYPE  MOUNTPOINTS
sda        8:0    0 59.6G  0 disk
├─sda1      8:1    0   620M  0 part  /boot/efi
├─sda2      8:2    0    5G    0 part
│ └─cr_swap 254:1    0    5G    0 crypt  [SWAP]
├─sda3      8:3    0   54G    0 part
│ └─cr_root 254:0    0   54G    0 crypt  /var
│                                     /srv
│                                     /usr/local
│                                     /opt
│                                     /home
│                                     /root
│                                     /
└─zd0       230:0    0   150M  0 disk
   └─md0     9:0    0    1G    0 raid5  /mnt/zstor/mock-data-md0
zd16       230:16  0   150M  0 disk
   └─md0     9:0    0    1G    0 raid5  /mnt/zstor/mock-data-md0
zd32       230:32  0   150M  0 disk
   └─md0     9:0    0    1G    0 raid5  /mnt/zstor/mock-data-md0
zd48       230:48  0   150M  0 disk
   └─md0     9:0    0    1G    0 raid5  /mnt/zstor/mock-data-md0
zd64       230:64  0   150M  0 disk
   └─md0     9:0    0    1G    0 raid5  /mnt/zstor/mock-data-md0
zd80       230:80  0   150M  0 disk
   └─md0     9:0    0    1G    0 raid5  /mnt/zstor/mock-data-md0
zd96       230:96  0   150M  0 disk
   └─md0     9:0    0    1G    0 raid5  /mnt/zstor/mock-data-md0
zd112      230:112  0   150M  0 disk
   └─md0     9:0    0    1G    0 raid5  /mnt/zstor/mock-data-md0
```

```
total 21G
512 drwxr-xr-x 2 root root 4 Jun 29 16:36 drz3-2hs-16d
512 drwxr-xr-x 2 root root 4 Jun 29 16:26 drz3-2hs-24d
0 drwxrwxrwx 2 root root 51 Jun 29 16:36 mock-data-md0
512 drwxr-xr-x 2 root root 4 Jun 29 16:38 rz3-0hs
512 drwxr-xr-x 2 root root 4 Jun 29 16:36 rz3-0hs-16d
512 drwxr-xr-x 2 root root 4 Jun 29 16:36 rz3-2hs-16d
512 drwxr-xr-x 2 root root 4 Jun 29 16:36 rz6-0hs
512 drwxr-xr-x 2 root root 4 Jun 29 16:36 rz6-1hs
20G -rwxrwxrwx. 1 root root 20G Jun 29 16:42 zstor.stor
512 drwxrwxrwx 2 root root 2 Jun 29 13:01 zvols
```

← Mock data model representation

```
total 401M
4.0K -rw-r--r--. 1 root root 3.3K Jun 29 14:06 pool-info.txt
400M -rw-r--r--. 1 root root 400M Jun 29 15:51 utilization.data
```

Mock data  
representation of  
existing output from  
the storage server

```
@ [redacted] :~$
total 32K
4.0K drwxr-x--- 9 [redacted] 4.0K Jun 25 20:22 [redacted]
4.0K drwxr-x--- 2 [redacted] 4.0K Mar 31 13:33 [redacted]
4.0K drwxr-x--- 4 [redacted] 4.0K Mar 25 08:21 [redacted]
4.0K drwxr-x--- 3 [redacted] 4.0K Feb 12 23:18 [redacted]
4.0K drwxr-x--- 4 [redacted] 4.0K Feb 12 03:29 [redacted]
4.0K drwxr-x--- 3 [redacted] 4.0K Jan 12 19:17 [redacted]
4.0K drwxr-x--- 6 [redacted] 4.0K Jan 12 14:29 [redacted]
```

~ estimate 40%  
utilization of total  
available storage

```
[redacted] :~$
Filesystem              Size  Used Avail Use% Mounted on
/dev/md0                 98T   39T   60T  39% /data
@ [redacted] ~$

xfs file system
```

Mock scale of  
current 8 disk @  
15TB in Raid 5 as  
shown around 100T  
with ~40T utilized  
and ~60T available



# Similar storage layout against actual existing environment

```
zd0      230:0      0 150M 0 disk /mnt/zstor/mock-data-md0
└md0     9:0      0 1G 0 raid5 /mnt/zstor/mock-data-md0
zd16     230:16     0 150M 0 disk /mnt/zstor/mock-data-md0
└md0     9:0      0 1G 0 raid5 /mnt/zstor/mock-data-md0
zd32     230:32     0 150M 0 disk /mnt/zstor/mock-data-md0
└md0     9:0      0 1G 0 raid5 /mnt/zstor/mock-data-md0
zd48     230:48     0 150M 0 disk /mnt/zstor/mock-data-md0
└md0     9:0      0 1G 0 raid5 /mnt/zstor/mock-data-md0
zd64     230:64     0 150M 0 disk /mnt/zstor/mock-data-md0
└md0     9:0      0 1G 0 raid5 /mnt/zstor/mock-data-md0
zd80     230:80     0 150M 0 disk /mnt/zstor/mock-data-md0
└md0     9:0      0 1G 0 raid5 /mnt/zstor/mock-data-md0
zd96     230:96     0 150M 0 disk /mnt/zstor/mock-data-md0
└md0     9:0      0 1G 0 raid5 /mnt/zstor/mock-data-md0
zd112    230:112    0 150M 0 disk /mnt/zstor/mock-data-md0
└md0     9:0      0 1G 0 raid5 /mnt/zstor/mock-data-md0
```

```
/dev/md0:
  Version : 1.2
  Creation Time : Mon Jun 29 13:49:33 2026
  Raid Level : raid5
  Array Size : 1060864 (1036.00 MiB 1086.32 MB)
  Used Dev Size : 151552 (148.00 MiB 155.19 MB)
  Raid Devices : 8
  Total Devices : 8
  Persistence : Superblock is persistent

  Intent Bitmap : Internal

  Update Time : Tue Jun 30 01:00:19 2026
  State : clean
  Active Devices : 8
  Working Devices : 8
  Failed Devices : 0
  Spare Devices : 0

  Layout : left-symmetric
  Chunk Size : 512K

  Consistency Policy : bitmap

  Name : k-nuc:0 (local to host k-nuc)
  UUID : 52410df6:532ae1ed:6b315a73:6aa727b1
  Events : 24
```

| Number | Major | Minor | RaidDevice | State  |      |             |
|--------|-------|-------|------------|--------|------|-------------|
| 0      | 230   | 272   | 0          | active | sync | /dev/zd272  |
| 1      | 230   | 1040  | 1          | active | sync | /dev/zd1040 |
| 2      | 230   | 1456  | 2          | active | sync | /dev/zd1456 |
| 3      | 230   | 736   | 3          | active | sync | /dev/zd736  |
| 4      | 230   | 624   | 4          | active | sync | /dev/zd624  |
| 5      | 230   | 1520  | 5          | active | sync | /dev/zd1520 |
| 6      | 230   | 1136  | 6          | active | sync | /dev/zd1136 |
| 8      | 230   | 96    | 7          | active | sync | /dev/zd96   |

# Actual existing environment

```
nvme0n1          259:0      0    14T  0 disk
??nvme0n1p1      259:2      0    14T  0 part
  ??md0          9:0       0   97.8T  0 raid5 /data
nvme1n1          259:1      0    14T  0 disk
??nvme1n1p1      259:4      0    14T  0 part
  ??md0          9:0       0   97.8T  0 raid5 /data
nvme5n1          259:3      0    14T  0 disk
??nvme5n1p1      259:5      0    14T  0 part
  ??md0          9:0       0   97.8T  0 raid5 /data
nvme2n1          259:6      0    14T  0 disk
??nvme2n1p1      259:7      0    14T  0 part
  ??md0          9:0       0   97.8T  0 raid5 /data
nvme3n1          259:8      0    14T  0 disk
??nvme3n1p1      259:9      0    14T  0 part
  ??md0          9:0       0   97.8T  0 raid5 /data
nvme4n1          259:10     0    14T  0 disk
??nvme4n1p1      259:12     0    14T  0 part
  ??md0          9:0       0   97.8T  0 raid5 /data
nvme6n1          259:11     0    14T  0 disk
??nvme6n1p1      259:13     0    14T  0 part
  ??md0          9:0       0   97.8T  0 raid5 /data
nvme7n1          259:14     0    14T  0 disk
??nvme7n1p1      259:15     0    14T  0 part
  ??md0          9:0       0   97.8T  0 raid5 /data
@llmstorage:~$
```

```
@llmstorage:~$ cat /dev/md0:
Version : 4.2
Creation Time: Fri Jan  9 16:41:18 2026
Raid Level: raid5
Array Size : 105005562368 (97.79 TiB 107.53 TB)
Used Dev Size : 15000794624 (13.97 TiB 15.36 TB)
Raid Devices : 8
Total Devices : 8
Persistence : Superblock is persistent

Intent Bitmap : Internal

Update Time : Thu Jun 25 20:25:00 2026
State : clean
Active Devices : 8
Working Devices : 8
Failed Devices : 0
Spare Devices : 0

Layout : left-symmetric
Chunk Size : 512K

Consistency Policy : bitmap

Name : @llmstorage:0 (local to host @llmstorage)
UUID : ed9de3f6:872b02d5:810f5111:cf836513
Events : 12374

Number Major Minor RaidDevice State
  0     259      2        0 active sync /dev/nvme0n1p1
  1     259      4        1 active sync /dev/nvme1n1p1
  2     259      7        2 active sync /dev/nvme2n1p1
  3     259      9        3 active sync /dev/nvme3n1p1
  4     259     12        4 active sync /dev/nvme4n1p1
  5     259      5        5 active sync /dev/nvme5n1p1
  6     259     13        6 active sync /dev/nvme6n1p1
  7     259     15        7 active sync /dev/nvme7n1p1
@llmstorage:~$
```



## Alternate storage layouts with more disk fault tolerance and resiliency

Raid 6 : 2 disk failures – data in critical state, 3<sup>rd</sup> failure means data loss

-same with 1 hot spare : time/notification saver for automated data rebuild

Raidz3 : 3 disk failures – data in critical state, 4<sup>th</sup> failure means data loss

-same with 2 hot spare : 2 disk will kick in for automated data rebuild, 1 still need intervention for data rebuild

-same with 16 disk layout (possible with current setup)

Declustered (distributed/draid) raidz3 : all disks in pool utilized for parity and hot spare worth specified, more beneficial on the data rebuild aspect

Draidz3 and the Raidz3 is recommended with 2 hot spares given number of disks in pool for data/parity ratio compromise

## Mock environment comparison output

| NAME         | SIZE  | ALLOC | FREE  | CKPOINT | EXPANDSZ | FRAG | CAP | DEDUP | HEALTH | ALROOT |
|--------------|-------|-------|-------|---------|----------|------|-----|-------|--------|--------|
| drz3-2hs-16d | 1.25G | 738M  | 540M  | -       | -        | 28%  | 57% | 1.00x | ONLINE | -      |
| drz3-2hs-24d | 2.00G | 1.13G | 890M  | -       | -        | 22%  | 56% | 1.00x | ONLINE | -      |
| rz3-0hs      | 1G    | 410M  | 614M  | -       | -        | 17%  | 40% | 1.00x | ONLINE | -      |
| rz3-0hs-16d  | 2G    | 1.18G | 838M  | -       | -        | 15%  | 59% | 1.00x | ONLINE | -      |
| rz3-2hs-16d  | 1.75G | 1.18G | 580M  | -       | -        | 22%  | 67% | 1.00x | ONLINE | -      |
| rz6-0hs      | 1G    | 607M  | 417M  | -       | -        | 19%  | 59% | 1.00x | ONLINE | -      |
| rz6-1hs      | 896M  | 608M  | 288M  | -       | -        | 29%  | 67% | 1.00x | ONLINE | -      |
| zvolumes     | 19.5G | 5.71G | 13.8G | -       | -        | 26%  | 29% | 1.00x | ONLINE | -      |

|              |      |      |      |     |                          |
|--------------|------|------|------|-----|--------------------------|
| zvolumes     | 2.2G | 128K | 2.2G | 1%  | /mnt/zstor/zvols         |
| /dev/md0     | 972M | 452M | 521M | 47% | /mnt/zstor/mock-data-md0 |
| rz6-0hs      | 551M | 401M | 150M | 73% | /mnt/zstor/rz6-0hs       |
| rz6-1hs      | 465M | 401M | 64M  | 87% | /mnt/zstor/rz6-1hs       |
| rz3-0hs      | 381M | 202M | 179M | 53% | /mnt/zstor/rz3-0hs       |
| rz3-0hs-16d  | 1.3G | 802M | 430M | 66% | /mnt/zstor/rz3-0hs-16d   |
| rz3-2hs-16d  | 1.1G | 802M | 259M | 76% | /mnt/zstor/rz3-2hs-16d   |
| drz3-2hs-16d | 584M | 404M | 180M | 70% | /mnt/zstor/drz3-2hs-16d  |
| drz3-2hs-24d | 600M | 405M | 196M | 68% | /mnt/zstor/drz3-2hs-24d  |

Raid 6 pool  
(8 disk layout),  
tolerant of 2 disk  
failures before data  
loss

```

pool: rz6-0hs
state: ONLINE
config:

    NAME                STATE                READ  WRITE  CKSUM
    rz6-0hs              ONLINE              0     0     0
      raidz2-0
        zvol/zvolumes/v008 ONLINE              0     0     0
        zvol/zvolumes/v009 ONLINE              0     0     0
        zvol/zvolumes/v010 ONLINE              0     0     0
        zvol/zvolumes/v011 ONLINE              0     0     0
        zvol/zvolumes/v012 ONLINE              0     0     0
        zvol/zvolumes/v013 ONLINE              0     0     0
        zvol/zvolumes/v014 ONLINE              0     0     0
        zvol/zvolumes/v015 ONLINE              0     0     0

errors: No known data errors

```

Raid 6 pool  
(8 disk layout)  
with a hot spare ,  
tolerant of 2 disk  
failures, with one  
active rebuild before  
data loss

```

pool: rz6-1hs
state: ONLINE
config:

    NAME                STATE                READ  WRITE  CKSUM
    rz6-1hs              ONLINE              0     0     0
      raidz2-0
        zvol/zvolumes/v016 ONLINE              0     0     0
        zvol/zvolumes/v017 ONLINE              0     0     0
        zvol/zvolumes/v018 ONLINE              0     0     0
        zvol/zvolumes/v019 ONLINE              0     0     0
        zvol/zvolumes/v020 ONLINE              0     0     0
        zvol/zvolumes/v021 ONLINE              0     0     0
        zvol/zvolumes/v022 ONLINE              0     0     0
      spares
        zvol/zvolumes/v023 AVAIL

errors: No known data errors

```

Raidz3 pool  
(8 disk layout),  
tolerant of 3 disk failures  
before data loss

```
pool: rz3-0hs
state: ONLINE
config:

    NAME                                STATE      READ  WRITE CKSUM
    rz3-0hs                             ONLINE      0     0     0
      raidz3-0                           ONLINE      0     0     0
        zvol/zvolumes/v024              ONLINE      0     0     0
        zvol/zvolumes/v025              ONLINE      0     0     0
        zvol/zvolumes/v026              ONLINE      0     0     0
        zvol/zvolumes/v027              ONLINE      0     0     0
        zvol/zvolumes/v028              ONLINE      0     0     0
        zvol/zvolumes/v029              ONLINE      0     0     0
        zvol/zvolumes/v030              ONLINE      0     0     0
        zvol/zvolumes/v031              ONLINE      0     0     0

errors: No known data errors
```

Raidz3 pool  
(16 disk layout),  
tolerant of 3 disk failures  
before data loss

```
pool: rz3-0hs-16d
state: ONLINE
config:

    NAME                                STATE      READ  WRITE CKSUM
    rz3-0hs-16d                         ONLINE      0     0     0
      raidz3-0                           ONLINE      0     0     0
        zvol/zvolumes/v032              ONLINE      0     0     0
        zvol/zvolumes/v033              ONLINE      0     0     0
        zvol/zvolumes/v034              ONLINE      0     0     0
        zvol/zvolumes/v035              ONLINE      0     0     0
        zvol/zvolumes/v036              ONLINE      0     0     0
        zvol/zvolumes/v037              ONLINE      0     0     0
        zvol/zvolumes/v038              ONLINE      0     0     0
        zvol/zvolumes/v039              ONLINE      0     0     0
        zvol/zvolumes/v040              ONLINE      0     0     0
        zvol/zvolumes/v041              ONLINE      0     0     0
        zvol/zvolumes/v042              ONLINE      0     0     0
        zvol/zvolumes/v043              ONLINE      0     0     0
        zvol/zvolumes/v044              ONLINE      0     0     0
        zvol/zvolumes/v045              ONLINE      0     0     0
        zvol/zvolumes/v046              ONLINE      0     0     0
        zvol/zvolumes/v047              ONLINE      0     0     0

errors: No known data errors
```



Raidz3 pool  
(16 disk layout),  
with 2 hot spares, tolerant  
of 3 disk failures, with two  
active rebuild before data  
loss

```
pool: rz3-2hs-16d
state: ONLINE
config:

    NAME                STATE        READ  WRITE CKSUM
    rz3-2hs-16d          ONLINE       0     0     0
    raidz3-0             ONLINE       0     0     0
    zvol/zvolumes/v048  ONLINE       0     0     0
    zvol/zvolumes/v049  ONLINE       0     0     0
    zvol/zvolumes/v050  ONLINE       0     0     0
    zvol/zvolumes/v051  ONLINE       0     0     0
    zvol/zvolumes/v052  ONLINE       0     0     0
    zvol/zvolumes/v053  ONLINE       0     0     0
    zvol/zvolumes/v054  ONLINE       0     0     0
    zvol/zvolumes/v055  ONLINE       0     0     0
    zvol/zvolumes/v056  ONLINE       0     0     0
    zvol/zvolumes/v057  ONLINE       0     0     0
    zvol/zvolumes/v058  ONLINE       0     0     0
    zvol/zvolumes/v059  ONLINE       0     0     0
    zvol/zvolumes/v060  ONLINE       0     0     0
    zvol/zvolumes/v061  ONLINE       0     0     0
    spares
    zvol/zvolumes/v062  AVAIL
    zvol/zvolumes/v063  AVAIL

errors: No known data errors
```

Declustered  
(distributed/draid) raidz3  
pool (16 disk layout),  
with 2 hot spares, tolerant  
of 3 disk failures, with two  
active rebuild before data  
loss

```
pool: drz3-2hs-16d
state: ONLINE
scan: resilvered 15.0M in 00:00:11 with 0 errors on Mon Jul 6 16:02:36 2026
config:

    NAME                STATE        READ  WRITE CKSUM
    drz3-2hs-16d         ONLINE       0     0     0
    draid3:11d:16c:2s-0  ONLINE       0     0     0
    zvol/zvolumes/v064  ONLINE       0     0     0
    zvol/zvolumes/v065  ONLINE       0     0     0
    zvol/zvolumes/v066  ONLINE       0     0     0
    zvol/zvolumes/v067  ONLINE       0     0     0
    zvol/zvolumes/v068  ONLINE       0     0     0
    zvol/zvolumes/v069  ONLINE       0     0     0
    zvol/zvolumes/v070  ONLINE       0     0     0
    zvol/zvolumes/v071  ONLINE       0     0     0
    zvol/zvolumes/v072  ONLINE       0     0     0
    zvol/zvolumes/v073  ONLINE       0     0     0
    zvol/zvolumes/v074  ONLINE       0     0     0
    zvol/zvolumes/v075  ONLINE       0     0     0
    zvol/zvolumes/v076  ONLINE       0     0     0
    zvol/zvolumes/v077  ONLINE       0     0     0
    zvol/zvolumes/v078  ONLINE       0     0     0
    zvol/zvolumes/v079  ONLINE       0     0     0
    spares
    draid3-0-0           AVAIL
    draid3-0-1           AVAIL

errors: No known data errors
```

Declustered  
(distributed/draid) raidz3  
pool (24 disk layout),  
with 2 hot spares, tolerant  
of 3 disk failures, with two  
active rebuild before data  
loss

```
pool: drz3-2hs-24d
state: ONLINE
config:
```

| NAME                | STATE  | READ | WRITE | CKSUM |
|---------------------|--------|------|-------|-------|
| drz3-2hs-24d        | ONLINE | 0    | 0     | 0     |
| draid3:19d:24c:2s-0 | ONLINE | 0    | 0     | 0     |
| zvol/zvolumes/v080  | ONLINE | 0    | 0     | 0     |
| zvol/zvolumes/v081  | ONLINE | 0    | 0     | 0     |
| zvol/zvolumes/v082  | ONLINE | 0    | 0     | 0     |
| zvol/zvolumes/v083  | ONLINE | 0    | 0     | 0     |
| zvol/zvolumes/v084  | ONLINE | 0    | 0     | 0     |
| zvol/zvolumes/v085  | ONLINE | 0    | 0     | 0     |
| zvol/zvolumes/v086  | ONLINE | 0    | 0     | 0     |
| zvol/zvolumes/v087  | ONLINE | 0    | 0     | 0     |
| zvol/zvolumes/v088  | ONLINE | 0    | 0     | 0     |
| zvol/zvolumes/v089  | ONLINE | 0    | 0     | 0     |
| zvol/zvolumes/v090  | ONLINE | 0    | 0     | 0     |
| zvol/zvolumes/v091  | ONLINE | 0    | 0     | 0     |
| zvol/zvolumes/v092  | ONLINE | 0    | 0     | 0     |
| zvol/zvolumes/v093  | ONLINE | 0    | 0     | 0     |
| zvol/zvolumes/v094  | ONLINE | 0    | 0     | 0     |
| zvol/zvolumes/v095  | ONLINE | 0    | 0     | 0     |
| zvol/zvolumes/v096  | ONLINE | 0    | 0     | 0     |
| zvol/zvolumes/v097  | ONLINE | 0    | 0     | 0     |
| zvol/zvolumes/v098  | ONLINE | 0    | 0     | 0     |
| zvol/zvolumes/v099  | ONLINE | 0    | 0     | 0     |
| zvol/zvolumes/v100  | ONLINE | 0    | 0     | 0     |
| zvol/zvolumes/v101  | ONLINE | 0    | 0     | 0     |
| zvol/zvolumes/v102  | ONLINE | 0    | 0     | 0     |
| zvol/zvolumes/v103  | ONLINE | 0    | 0     | 0     |
| spares              |        |      |       |       |
| draid3-0-0          | AVAIL  |      |       |       |
| draid3-0-1          | AVAIL  |      |       |       |

```
errors: No known data errors
```



# Disk Failure Simulation – Mock Environment

1 disk failure on the raid5 pool, tolerant of 1 disk failures

```
/dev/md0:
  Version : 1.2
  Creation Time : Mon Jun 29 13:49:33 2026
  Raid Level : raid5
  Array Size : 1060864 (1036.00 MiB 1086.32 MB)
  Used Dev Size : 151552 (148.00 MiB 155.1 MB)
  Raid Devices : 8
  Total Devices : 8
  Persistence : Superblock is persistent

  Intent Bitmap : Internal

  Update Time : Wed Jul 1 13:58:07 2026
  State : clean, degraded
  Active Devices : 7
  Working Devices : 7
  Failed Devices : 1
  Spare Devices : 0

  Layout : left-symmetric
  Chunk Size : 512K
  Consistency Policy : bitmap

  Name : k-nuc:0 (local to host k-nuc)
  UUID : 52410df6:532ae1ed:6b315a73:6aa727b1
  Events : 26

Number   Major   Minor   RaidDevice State
- - - - -
  0         0         0         0   removed
  1        230        1040        1   active sync  /dev/zd1040
  2        230        1456        2   active sync  /dev/zd1456
  3        230         736        3   active sync  /dev/zd736
  4        230         624        4   active sync  /dev/zd624
  5        230        1520        5   active sync  /dev/zd1520
  6        230        1136        6   active sync  /dev/zd1136
  7        230         96         7   active sync  /dev/zd96
  0        230        272        -   faulty  /dev/zd272
```

# Disk Failure Simulation – Mock Environment

1 disk failure on the raid5 pool, tolerant of 1 disk failures

```
kernel: md: recover of RAID array md0
mdadm[28691]: mdadm: NewArray event detected on md device /dev/md0
mdadm[28691]: mdadm: DegradedArray event detected on md device /dev/md0
mdadm[28691]: mdadm: RebuildStarted event detected on md device /dev/md0
kernel: md: md0: recover done.
mdadm[28691]: mdadm: RebuildFinished event detected on md device /dev/md0
mdadm[28691]: mdadm: SpareActive event detected on md device /dev/md0, component device /dev/zd2032
mdadm[28691]: mdadm: DeviceDisappeared event detected on md device /dev/md0
mdadm[28691]: mdadm: error opening /dev/md0: No such device
blkdeactivate[32690]: [MD]: deactivating raid5 device md0...
kernel: md/raid:md0: device zd96 operational as raid disk 6
kernel: md/raid:md0: device zd80 operational as raid disk 5
kernel: md/raid:md0: device zd64 operational as raid disk 4
kernel: md/raid:md0: device zd48 operational as raid disk 3
kernel: md/raid:md0: device zd32 operational as raid disk 2
kernel: md/raid:md0: device zd16 operational as raid disk 1
kernel: md/raid:md0: device zd0 operational as raid disk 0
kernel: md/raid:md0: raid level 5 active with 7 out of 8 devices, algorithm 2
kernel: md0: array will not be assembled in old kernels that lack configurable LBS support (<= 6.18)
kernel: md0: detected capacity change from 0 to 2121728
kernel: md: recover of RAID array md0
```



```

Personalities : [raid4] [raid5] [raid6]
md0 : broken raid5 zd272[0](F) zd1520[5] zd1456[2] zd1136[6] zd1040[1](F) zd736[3] zd624[4] zd96[8]
      1060864 blocks super 1.2 level 5, 512k chunk, algorithm 2 [8/6] [___UUUUUU]
      bitmap: 0/1 pages [0KB], 65536KB chunk

unused devices: <none>

```

```

kernel: md/raid:md0: Disk failure on zd272, disabling device.
kernel: md/raid:md0: Operation continuing on 7 devices.
kernel: md/raid:md0: Disk failure on zd1040, disabling device.
kernel: md/raid:md0: Cannot continue operation (2/8 failed).
mdadm[8576]: mdadm: Fail event detected on md device /dev/md0, component device /dev/zd272
mdadm[8576]: mdadm: Fail event detected on md device /dev/md0, component device /dev/zd1040

```

```

/dev/md0:
  Version : 1.2
  Creation Time : Mon Jun 29 13:49:33 2026
  Raid Level : raid5
  Array Size : 1060864 (1036.00 MiB 1086.32 MB)
  Used Dev Size : 151552 (148.00 MiB 155.19 MB)
  Raid Devices : 8
  Total Devices : 8
  Persistence : Superblock is persistent

  Intent Bitmap : Internal

  Update Time : Wed Jul 1 14:54:43 2026
  State : clean, FAILED
  Active Devices : 6
  Working Devices : 6
  Failed Devices : 2
  Spare Devices : 0

  Layout : left-symmetric
  Chunk Size : 512K

Consistency Policy : bitmap

  Name : k-nuc:0 (local to host k-nuc)
  UUID : 52410df6:532ae1ed:6a15a73:6aa727b1
  Events : 38

```

| Number | Major | Minor | RaidDevice | State       | Device      |
|--------|-------|-------|------------|-------------|-------------|
| -      | 0     | 0     | 0          | removed     |             |
| -      | 0     | 0     | 1          | removed     |             |
| 2      | 230   | 1456  | 2          | active sync | /dev/zd1456 |
| 3      | 230   | 736   | 3          | active sync | /dev/zd736  |
| 4      | 230   | 624   | 4          | active sync | /dev/zd624  |
| 5      | 230   | 1520  | 5          | active sync | /dev/zd1520 |
| 6      | 230   | 1136  | 6          | active sync | /dev/zd1136 |
| 8      | 230   | 96    | 7          | active sync | /dev/zd96   |
| 0      | 230   | 272   | -          | faulty      | /dev/zd272  |
| 1      | 230   | 1040  | -          | faulty      | /dev/zd1040 |

# Disk Failure Simulation – Mock Environment

2 disk failure on the raid5 pool,  
tolerant of 1 disk failures

## Data Loss

System  
forcefully  
dismounted,  
no data available  
at mount point

```

root root 4 Jun 29 16:36 drz3-2hs-16d
root root 4 Jun 29 16:26 drz3-2hs-24d
root root 0 Jun 29 14:02 mock-data-md0
root root 4 Jun 29 16:38 rz3-0hs
root root 4 Jun 29 16:36 rz3-0hs-16d
root root 4 Jun 29 16:36 rz3-2hs-16d
root root 0 Jun 29 14:40 rz6-0hs
root root 4 Jun 29 16:36 rz6-1hs
root root 20G Jul 1 15:00 zstor.stor
root root 2 Jun 29 13:01 zvols

```

```

# ls -lsh /mnt/zstor/mock-data-md0/
total 0

```



# Disks Failure – Live Environment

Prior example of raid5 (1 disk failure), this is real failure of raid6 (2 disk failure)

```
md0 : active raid6 nvme7n1p1[7] nvme6n1p1[6] nvme5n1p1[5]
nvme4n1p1[4] nvme3n1p1[3] nvme2n1p1[2] nvme1n1p1[1] nvme0n1p1[0]
11251445760 blocks super 1.2 level 6, 512k chunk, algorithm 2 [8/8]
```

```
[UUUUUUUU]
```

```
bitmap: 0/14 pages [0KB], 65536KB chunk
```

```
/dev/md0:
```

```
Version : 1.2
```

```
Creation Time : Tue Apr 28 00:53:14 2026
```

```
Raid Level : raid6
```

```
Array Size : 11251445760 (10.48 TiB 11.52 TB)
```

```
Used Dev Size : 1875240960 (1788.37 GiB 1920.25 GB)
```

```
Raid Devices : 8
```

```
Total Devices : 8
```

```
Persistence : Superblock is persistent
```

```
Intent Bitmap : Internal
```

```
Update Time : Tue Apr 28 03:29:40 2026
```

```
State : clean
```

```
Active Devices : 8
```

```
Working Devices : 8
```

```
Failed Devices : 0
```

```
Spare Devices : 0
```

```
kernel: md/raid:md0: Disk failure on nvme2n1p1, disabling device.
```

```
kernel: md/raid:md0: Operation continuing on 7 devices.
```

```
md/raid:md0: Disk failure on nvme3n1p1, disabling device.
```

```
md/raid:md0: Operation continuing on 6 devices.
```

```
md/raid:md0: Disk failure on nvme0n1p1, disabling device.
```

```
md/raid:md0: Cannot continue operation (3/8 failed).
```

```
md/raid:md0: Disk failure on nvme6n1p1, disabling device.
```

```
md/raid:md0: Cannot continue operation (4/8 failed).
```

```
md/raid:md0: Disk failure on nvme1n1p1, disabling device.
```

```
md/raid:md0: Cannot continue operation (5/8 failed).
```

```
md/raid:md0: Disk failure on nvme7n1p1, disabling device.
```

```
md/raid:md0: Cannot continue operation (6/8 failed).
```

```
/dev/md0:
```

```
Version : 1.2
```

```
Raid Level : raid6
```

```
Total Devices : 2
```

```
Persistence : Superblock is persistent
```

```
State : inactive
```

```
Working Devices : 2
```

```
Name :
```

```
UUID :
```

```
dc6a9da3:aca8d91a:247568ed:71042b22
```

```
Events : 1816
```

```
Number Major Minor RaidDevice
```

```
- 259 2 - /dev/nvme0n1p1
```

```
- 259 5 - /dev/nvme1n1p1
```

2 disks failure at the same time is statistically less likely

6 disks failure at the same time is statistically improbable, but as evidenced, not impossible

~10 Terabyte pool failed, multiple terabytes of data lost (unless backed up)

# Disk Failure Simulation – Mock Environment

3 disk failures on the raid6 pool, tolerant of 2 disk failures

```
spa.c:7553:spa_tryimport(): spa_tryimport: importing rz6-0hs  
spa_misc.c:431:spa_load_note(): spa_load(rz6-0hs, config trusted): LOADING
```

```
1782907539 ffff8db676571ac0 spa_misc.c:431:spa_load_note(): spa_load(rz6-0hs, c  
1782907539 ffff8db676571ac0 spa_misc.c:431:spa_load_note(): spa_load(rz6-0hs, c  
op-level vdevs at this stage.  
1782907539 ffff8db676571ac0 spa_misc.c:417:spa_load_failed(): spa_load(rz6-0hs,  
1782907539 ffff8db676571ac0 vdev.c:237:vdev_dbgmsg_print_tree(): vdev 0: root  
1782907539 ffff8db676571ac0 vdev.c:237:vdev_dbgmsg_print_tree(): vdev 0: ra  
1782907539 ffff8db676571ac0 vdev.c:237:vdev_dbgmsg_print_tree(): vdev 0:  
part1, can't open  
1782907539 ffff8db676571ac0 vdev.c:237:vdev_dbgmsg_print_tree(): vdev 1:  
art1, can't open  
1782907539 ffff8db676571ac0 vdev.c:237:vdev_dbgmsg_print_tree(): vdev 2:  
art1, can't open
```

→ cannot import 'rz6-0hs': no such pool or dataset  
Destroy and re-create the pool from  
a backup source.

```
pool: rz6-0hs  
id: 9332855663710121757  
state: UNAVAIL  
status: One or more devices contains corrupted data.  
action: The pool cannot be imported due to damaged devices or data.  
see: https://openzfs.github.io/openzfs-docs/msg/ZFS-8000-5E  
config:
```

|                    |         |                       |
|--------------------|---------|-----------------------|
| rz6-0hs            | UNAVAIL | insufficient replicas |
| raidz2-0           | UNAVAIL | insufficient replicas |
| zvol/zvolumes/v008 | UNAVAIL |                       |
| zvol/zvolumes/v009 | UNAVAIL |                       |
| zvol/zvolumes/v010 | UNAVAIL |                       |
| zvol/zvolumes/v011 | ONLINE  |                       |
| zvol/zvolumes/v012 | ONLINE  |                       |
| zvol/zvolumes/v013 | ONLINE  |                       |
| zvol/zvolumes/v014 | ONLINE  |                       |
| zvol/zvolumes/v015 | ONLINE  |                       |

# Disk Failure Simulation

## – Mock Environment

Healthy Pool  
to Degraded

1 Disk Failure, kicked off  
Spare to rebuild (resilver)

Spare status shows avail  
due to speed of resilver,  
with limited mock data

```
pool: rz6-1hs
state: ONLINE
config:

    NAME                STATE        READ  WRITE CKSUM
    rz6-1hs              ONLINE       0     0     0
    raidz2-0             ONLINE       0     0     0
        zvol/zvolumes/v016 ONLINE       0     0     0
        zvol/zvolumes/v017 ONLINE       0     0     0
        zvol/zvolumes/v018 ONLINE       0     0     0
        zvol/zvolumes/v019 ONLINE       0     0     0
        zvol/zvolumes/v020 ONLINE       0     0     0
        zvol/zvolumes/v021 ONLINE       0     0     0
        zvol/zvolumes/v022 ONLINE       0     0     0
    spares
        zvol/zvolumes/v023 AVAIL

errors: No known data errors
```

```
pool: rz6-1hs
state: DEGRADED
status: One or more devices has been taken offline by the administrator.
        Sufficient replicas exist for the pool to continue functioning in a
        degraded state.
action: Online the device using 'zpool online' or replace the device with
        'zpool replace'.
scan: resilvered 0B in 00:00:01 with 0 errors on Mon Jul  6 10:05:18 2026
config:

    NAME                STATE        READ  WRITE CKSUM
    rz6-1hs              DEGRADED     0     0     0
    raidz2-0             DEGRADED     0     0     0
        zvol/zvolumes/v016 OFFLINE       0     0     0
        zvol/zvolumes/v017 ONLINE       0     0     0
        zvol/zvolumes/v018 ONLINE       0     0     0
        zvol/zvolumes/v019 ONLINE       0     0     0
        zvol/zvolumes/v020 ONLINE       0     0     0
        zvol/zvolumes/v021 ONLINE       0     0     0
        zvol/zvolumes/v022 ONLINE       0     0     0
    spares
        zvol/zvolumes/v023 AVAIL

errors: No known data errors
```

```
Jul  6 2026 10:05:17.999608583 sysevent.fs.zfs.resilver_start
version = 0x0
class = "sysevent.fs.zfs.resilver_start"
pool = "rz6-1hs"
pool_guid = 0x1e0229c176ab9a2f
pool_state = 0x0
pool_context = 0x0
resilver_type = "healing"
time = 0x6a4bb61d 0x3b94d107
eid = 0x2798
```

```
Jul  6 2026 10:05:18.320611022 sysevent.fs.zfs.resilver_finish
version = 0x0
class = "sysevent.fs.zfs.resilver_finish"
pool = "rz6-1hs"
pool_guid = 0x1e0229c176ab9a2f
pool_state = 0x0
pool_context = 0x0
resilver_type = "healing"
time = 0x6a4bb61e 0x131c22ce
eid = 0x279b
```



# Disk Failure Simulation

## – Mock Environment

Degraded pool, induced  
another disk failure

```
pool: rz6-1hs
state: DEGRADED
status: One or more devices has been taken offline by the administrator.
        Sufficient replicas exist for the pool to continue functioning in a
        degraded state.
action: Online the device using 'zpool online' or replace the device with
        'zpool replace'.
scan: resilvered 0B in 00:00:01 with 0 errors on Mon Jul  6 10:05:18 2026
config:

        NAME                                STATE      READ  WRITE CKSUM
        rz6-1hs                             DEGRADED   0     0     0
          raidz2-0                           DEGRADED   0     0     0
            zvol/zvolumes/v016              OFFLINE    0     0     0
            zvol/zvolumes/v017              ONLINE    0     0     0
            zvol/zvolumes/v018              ONLINE    0     0     0
            zvol/zvolumes/v019              ONLINE    0     0     0
            zvol/zvolumes/v020              ONLINE    0     0     0
            zvol/zvolumes/v021              ONLINE    0     0     0
            zvol/zvolumes/v022              ONLINE    0     0     0
        spares
          zvol/zvolumes/v023                AVAIL

errors: no known data errors
```

```
cannot import 'rz6-1hs': no such pool or dataset
Destroy and re-create the pool from
a backup source.
```

```
cannot import 'rz6-0hs': no such pool or dataset
Destroy and re-create the pool from
a backup source
```

```
Jul  6 2026 13:59:22.151276428 ereport.fs.zfs.vdev.no_replicas
class = "ereport.fs.zfs.vdev.no_replicas"
ena = 0x77240e64a4400001
detector = (embedded nvlist)
  version = 0x0
  scheme = "zfs"
  pool = 0x1e0229c176ab9a2f
  vdev = 0x413e1f369e0131e1
(end detector)
pool = "rz6-1hs"
pool_guid = 0x1e0229c176ab9a2f
```

```
Jul 06 11:37:42 k-nuc kernel: zio pool=rz6-1hs vdev=/dev/zvol/zvolumes/v018-part1 error=5 type=2 offset=65830912 size=16384
Jul 06 11:37:42 k-nuc kernel: zio pool=rz6-1hs vdev=/dev/zvol/zvolumes/v018-part1 error=5 type=2 offset=129581056 size=1638
Jul 06 11:37:42 k-nuc kernel: zio pool=rz6-1hs vdev=/dev/zvol/zvolumes/v018-part1 error=5 type=2 offset=18759680 size=16384
Jul 06 11:37:42 k-nuc kernel: zio pool=rz6-1hs vdev=/dev/zvol/zvolumes/v018-part1 error=5 type=1 offset=270336 size=8192 fl
Jul 06 11:37:42 k-nuc kernel: zio pool=rz6-1hs vdev=/dev/zvol/zvolumes/v018-part1 error=5 type=1 offset=146284544 size=8192
Jul 06 11:37:42 k-nuc kernel: zio pool=rz6-1hs vdev=/dev/zvol/zvolumes/v018-part1 error=5 type=1 offset=146546688 size=819
Jul 06 11:37:42 k-nuc kernel: WARNING: Pool 'rz6-1hs' has encountered an uncorrectable I/O failure and has been suspended.
Jul 06 11:39:35 k-nuc systemd[1]: Unmounting /mnt/zstor/rz6-1hs...
```

# Disk Failure Simulation

## – Mock Environment

```
pool: rz3-0hs
state: ONLINE
config:

    NAME                                STATE      READ  WRITE CKSUM
    rz3-0hs                             ONLINE         0     0     0
      raidz3-0                          ONLINE         0     0     0
        zvol/zvolumes/v024             ONLINE         0     0     0
        zvol/zvolumes/v025             ONLINE         0     0     0
        zvol/zvolumes/v026             ONLINE         0     0     0
        zvol/zvolumes/v027             ONLINE         0     0     0
        zvol/zvolumes/v028             ONLINE         0     0     0
        zvol/zvolumes/v029             ONLINE         0     0     0
        zvol/zvolumes/v030             ONLINE         0     0     0
        zvol/zvolumes/v031             ONLINE         0     0     0

errors: No known data errors
```

System is degraded with 3 disk failure/faults  
but still operational, state is critical

```
pool: rz3-0hs
state: DEGRADED
status: One or more devices has been taken offline by the administrator.
        Sufficient replicas exist for the pool to continue functioning in a
        degraded state.
action: Online the device using 'zpool online' or replace the device with
        'zpool replace'.
config:

    NAME                                STATE      READ  WRITE CKSUM
    rz3-0hs                             DEGRADED         0     0     0
      raidz3-0                          DEGRADED         0     0     0
        zvol/zvolumes/v024             OFFLINE         0     0     0
        zvol/zvolumes/v025             OFFLINE         0     0     0
        zvol/zvolumes/v026             OFFLINE         0     0     0
        zvol/zvolumes/v027             ONLINE         0     0     0
        zvol/zvolumes/v028             ONLINE         0     0     0
        zvol/zvolumes/v029             ONLINE         0     0     0
        zvol/zvolumes/v030             ONLINE         0     0     0
        zvol/zvolumes/v031             ONLINE         0     0     0

errors: No known data errors
```



# Disk Failure Simulation – Mock Environment

# Data scrubbing for self repair after disk faults cleared

```
pool: rz3-0hs
state: ONLINE
  scan: scrub in progress since Mon Jul 6 14:32:46 2026
        410M / 413M scanned, 0B / 410M issued
        0B repaired, 0.00% done, no estimated completion time
config:

      NAME                                STATE      READ WRITE CKSUM
      rz3-0hs                            ONLINE      0     0     0
        raidz3-0
          zvol/zvolumes/v024            ONLINE      0     0     0
          zvol/zvolumes/v025            ONLINE      0     0     0
          zvol/zvolumes/v026            ONLINE      0     0     0
          zvol/zvolumes/v027            ONLINE      0     0     0
          zvol/zvolumes/v028            ONLINE      0     0     0
          zvol/zvolumes/v029            ONLINE      0     0     0
          zvol/zvolumes/v030            ONLINE      0     0     0
          zvol/zvolumes/v031            ONLINE      0     0     0

errors: No known data errors
```

```
pool: rz3-0hs
state: ONLINE
scan: scrub repaired 0B in 00:00:12 with 0 errors on Mon Jul 6 14:32:58 2026
config:

    NAME                                STATE        READ  WRITE CKSUM
    rz3-0hs                             ONLINE       0     0     0
    raidz3-0                            ONLINE       0     0     0
```



# Disk Failure Simulation – Mock Environment

Note the 4 disk failures/faults and no mention of sufficient replicas exists to continue in degraded state

```
pool: rz3-0hs
state: DEGRADED
status: One or more devices has experienced an unrecoverable error. An
        attempt was made to correct the error. Applications are unaffected.
action: Determine if the device needs to be replaced, and clear the errors
        using 'zpool clear' or replace the device with 'zpool replace'.
        see: https://openzfs.github.io/openzfs-docs/msg/ZFS-8000-9P
scan: scrub repaired 0B in 00:00:12 with 0 errors on Mon Jul  6 14:32:58 2026
config:
```

| NAME               | STATE    | READ | WRITE | CKSUM |                 |
|--------------------|----------|------|-------|-------|-----------------|
| rz3-0hs            | DEGRADED | 0    | 0     | 0     |                 |
| raidz3-0           | DEGRADED | 0    | 0     | 0     |                 |
| zvol/zvolumes/v024 | DEGRADED | 0    | 0     | 0     | too many errors |
| zvol/zvolumes/v025 | DEGRADED | 0    | 0     | 0     | too many errors |
| zvol/zvolumes/v026 | DEGRADED | 0    | 0     | 0     | too many errors |
| zvol/zvolumes/v027 | DEGRADED | 0    | 0     | 0     | too many errors |
| zvol/zvolumes/v028 | ONLINE   | 0    | 0     | 0     |                 |
| zvol/zvolumes/v029 | ONLINE   | 0    | 0     | 0     |                 |
| zvol/zvolumes/v030 | ONLINE   | 0    | 0     | 0     |                 |
| zvol/zvolumes/v031 | ONLINE   | 0    | 0     | 0     |                 |

```
errors: No known data errors
```

# Disk Failure Simulation

## – Mock Environment

Healthy Pool  
to Degraded

```
pool: rz3-0hs-16d
state: ONLINE
config:

    NAME                                STATE      READ  WRITE CKSUM
    rz3-0hs-16d                        ONLINE         0     0     0
      raidz3-0
        zvol/zvolumes/v032           ONLINE         0     0     0
        zvol/zvolumes/v033           ONLINE         0     0     0
        zvol/zvolumes/v034           ONLINE         0     0     0
        zvol/zvolumes/v035           ONLINE         0     0     0
        zvol/zvolumes/v036           ONLINE         0     0     0
        zvol/zvolumes/v037           ONLINE         0     0     0
        zvol/zvolumes/v038           ONLINE         0     0     0
        zvol/zvolumes/v039           ONLINE         0     0     0
        zvol/zvolumes/v040           ONLINE         0     0     0
        zvol/zvolumes/v041           ONLINE         0     0     0
        zvol/zvolumes/v042           ONLINE         0     0     0
        zvol/zvolumes/v043           ONLINE         0     0     0
        zvol/zvolumes/v044           ONLINE         0     0     0
        zvol/zvolumes/v045           ONLINE         0     0     0
        zvol/zvolumes/v046           ONLINE         0     0     0
        zvol/zvolumes/v047           ONLINE         0     0     0

errors: No known data errors
```

System is degraded with 3 disk  
failure/faults but still  
operational, state is critical

```
pool: rz3-0hs-16d
state: DEGRADED
status: One or more devices has been taken offline by the administrator.
        Sufficient replicas exist for the pool to continue functioning in a
        degraded state.
action: online the device using 'zpool online' or replace the device with
        'zpool replace'.
config:

    NAME                                STATE      READ  WRITE CKSUM
    rz3-0hs-16d                        DEGRADED         0     0     0
      raidz3-0
        zvol/zvolumes/v032           OFFLINE         0     0     0
        zvol/zvolumes/v033           OFFLINE         0     0     0
        zvol/zvolumes/v034           OFFLINE         0     0     0
        zvol/zvolumes/v035           ONLINE         0     0     0
        zvol/zvolumes/v036           ONLINE         0     0     0
        zvol/zvolumes/v037           ONLINE         0     0     0
        zvol/zvolumes/v038           ONLINE         0     0     0
        zvol/zvolumes/v039           ONLINE         0     0     0
        zvol/zvolumes/v040           ONLINE         0     0     0
        zvol/zvolumes/v041           ONLINE         0     0     0
        zvol/zvolumes/v042           ONLINE         0     0     0
        zvol/zvolumes/v043           ONLINE         0     0     0
        zvol/zvolumes/v044           ONLINE         0     0     0
        zvol/zvolumes/v045           ONLINE         0     0     0
        zvol/zvolumes/v046           ONLINE         0     0     0
        zvol/zvolumes/v047           ONLINE         0     0     0

errors: No known data errors
```



# Disk Failure Simulation

## – Mock Environment

Degraded pool, induced  
another disk failure

```
pool: rz3-0hs-16d
state: DEGRADED
status: One or more devices has been taken offline by the administrator.
        Sufficient replicas exist for the pool to continue functioning in a
        degraded state.
action: Online the device using 'zpool online' or replace the device with
        'zpool replace'.
config:

        NAME                       STATE      READ  WRITE CKSUM
        rz3-0hs-16d                 DEGRADED   0     0     0
          raidz3-0
            zvol/zvolumes/v032      OFFLINE   0     0     0
            zvol/zvolumes/v033      OFFLINE   0     0     0
            zvol/zvolumes/v034      OFFLINE   0     0     0
            zvol/zvolumes/v035      ONLINE    0     0     0
            zvol/zvolumes/v036      ONLINE    0     0     0
            zvol/zvolumes/v037      ONLINE    0     0     0
            zvol/zvolumes/v038      ONLINE    0     0     0
            zvol/zvolumes/v039      ONLINE    0     0     0
            zvol/zvolumes/v040      ONLINE    0     0     0
            zvol/zvolumes/v041      ONLINE    0     0     0
            zvol/zvolumes/v042      ONLINE    0     0     0
            zvol/zvolumes/v043      ONLINE    0     0     0
            zvol/zvolumes/v044      ONLINE    0     0     0
            zvol/zvolumes/v045      ONLINE    0     0     0
            zvol/zvolumes/v046      ONLINE    0     0     0
            zvol/zvolumes/v047      ONLINE    0     0     0

errors: No known data errors
```

```
pool: rz3-0hs-16d
state: DEGRADED
status: One or more devices has experienced an unrecoverable error. An
        attempt was made to correct the error. Applications are unaffected.
action: Determine if the device needs to be replaced, and clear the errors
        using 'zpool clear' or replace the device with 'zpool replace'.
        see: https://openzfs.github.io/openzfs-docs/msg/ZFS-8000-9P
scan: scrub repaired 0B in 00:00:17 with 0 errors on Mon Jul  6 16:16:27 2026
config:
```

| NAME               | STATE    | READ | WRITE | CKSUM |
|--------------------|----------|------|-------|-------|
| rz3-0hs-16d        | DEGRADED | 0    | 0     | 0     |
| raidz3-0           | DEGRADED | 0    | 0     | 0     |
| zvol/zvolumes/v032 | DEGRADED | 0    | 0     | 0     |
| zvol/zvolumes/v033 | DEGRADED | 0    | 0     | 0     |
| zvol/zvolumes/v034 | DEGRADED | 0    | 0     | 0     |
| zvol/zvolumes/v035 | DEGRADED | 0    | 0     | 0     |
| zvol/zvolumes/v036 | ONLINE   | 0    | 0     | 0     |
| zvol/zvolumes/v037 | ONLINE   | 0    | 0     | 0     |
| zvol/zvolumes/v038 | ONLINE   | 0    | 0     | 0     |
| zvol/zvolumes/v039 | ONLINE   | 0    | 0     | 0     |
| zvol/zvolumes/v040 | ONLINE   | 0    | 0     | 0     |
| zvol/zvolumes/v041 | ONLINE   | 0    | 0     | 0     |
| zvol/zvolumes/v042 | ONLINE   | 0    | 0     | 0     |
| zvol/zvolumes/v043 | ONLINE   | 0    | 0     | 0     |
| zvol/zvolumes/v044 | ONLINE   | 0    | 0     | 0     |
| zvol/zvolumes/v045 | ONLINE   | 0    | 0     | 0     |
| zvol/zvolumes/v046 | ONLINE   | 0    | 0     | 0     |
| zvol/zvolumes/v047 | ONLINE   | 0    | 0     | 0     |

errors: No known data errors

Note the 4 disk  
failures/faults and no  
mention of sufficient  
replicas exists to  
continue in degraded  
state



# Disk Failure Simulation

## – Mock Environment

Healthy Pool to Degraded

Pool scrubbing/resilvering for data resiliency, even with the disks fault

```
dsl_scan_restart_resilver(): restarting resilver for rz3-0hs-16d at txg=108283
dsl_scan_restart_resilver(): restarting resilver for rz3-0hs-16d at txg=108286
dsl_scan_restart_resilver(): restarting resilver for rz3-0hs-16d at txg=108289
```

```
pool: rz3-2hs-16d
state: DEGRADED
status: One or more devices has been taken offline by the administrator.
Sufficient replicas exist for the pool to continue functioning in a degraded state.
action: Online the device using 'zpool online' or replace the device with 'zpool replace'.
config:
```

| NAME               | STATE    | READ | WRITE | CKSUM |
|--------------------|----------|------|-------|-------|
| rz3-2hs-16d        | DEGRADED | 0    | 0     | 0     |
| raidz3-0           | DEGRADED | 0    | 0     | 0     |
| zvol/zvolumes/v048 | OFFLINE  | 0    | 0     | 0     |
| zvol/zvolumes/v049 | OFFLINE  | 0    | 0     | 0     |
| zvol/zvolumes/v050 | OFFLINE  | 0    | 0     | 0     |
| zvol/zvolumes/v051 | ONLINE   | 0    | 0     | 0     |
| zvol/zvolumes/v052 | ONLINE   | 0    | 0     | 0     |
| zvol/zvolumes/v053 | ONLINE   | 0    | 0     | 0     |
| zvol/zvolumes/v054 | ONLINE   | 0    | 0     | 0     |
| zvol/zvolumes/v055 | ONLINE   | 0    | 0     | 0     |
| zvol/zvolumes/v056 | ONLINE   | 0    | 0     | 0     |
| zvol/zvolumes/v057 | ONLINE   | 0    | 0     | 0     |
| zvol/zvolumes/v058 | ONLINE   | 0    | 0     | 0     |
| zvol/zvolumes/v059 | ONLINE   | 0    | 0     | 0     |
| zvol/zvolumes/v060 | ONLINE   | 0    | 0     | 0     |
| zvol/zvolumes/v061 | ONLINE   | 0    | 0     | 0     |

```
spares
zvol/zvolumes/v062  AVAIL
zvol/zvolumes/v063  AVAIL
```

errors: No known data errors

```
pool: rz3-2hs-16d
state: ONLINE
config:
```

| NAME               | STATE  | READ | WRITE | CKSUM |
|--------------------|--------|------|-------|-------|
| rz3-2hs-16d        | ONLINE | 0    | 0     | 0     |
| raidz3-0           | ONLINE | 0    | 0     | 0     |
| zvol/zvolumes/v048 | ONLINE | 0    | 0     | 0     |
| zvol/zvolumes/v049 | ONLINE | 0    | 0     | 0     |
| zvol/zvolumes/v050 | ONLINE | 0    | 0     | 0     |
| zvol/zvolumes/v051 | ONLINE | 0    | 0     | 0     |
| zvol/zvolumes/v052 | ONLINE | 0    | 0     | 0     |
| zvol/zvolumes/v053 | ONLINE | 0    | 0     | 0     |
| zvol/zvolumes/v054 | ONLINE | 0    | 0     | 0     |
| zvol/zvolumes/v055 | ONLINE | 0    | 0     | 0     |
| zvol/zvolumes/v056 | ONLINE | 0    | 0     | 0     |
| zvol/zvolumes/v057 | ONLINE | 0    | 0     | 0     |
| zvol/zvolumes/v058 | ONLINE | 0    | 0     | 0     |
| zvol/zvolumes/v059 | ONLINE | 0    | 0     | 0     |
| zvol/zvolumes/v060 | ONLINE | 0    | 0     | 0     |
| zvol/zvolumes/v061 | ONLINE | 0    | 0     | 0     |

```
spares
zvol/zvolumes/v062  AVAIL
zvol/zvolumes/v063  AVAIL

errors: No known data errors
```

```
Jul  6 2026 16:53:26.389140052 sysevent.fs.zfs.scrub_finish
version = 0x0
class = "sysevent.fs.zfs.scrub_finish"
pool = "rz3-2hs-16d"
pool_guid = 0xae250084c8cfa1e
pool_state = 0x0
pool_context = 0x0
time = 0x6a4c15c6 0x1731ce54
eid = 0xd8
```

```
Jul  6 2026 16:53:11.179996777 sysevent.fs.zfs.scrub_start
version = 0x0
class = "sysevent.fs.zfs.scrub_start"
pool = "rz3-2hs-16d"
pool_guid = 0xae250084c8cfa1e
pool_state = 0x0
pool_context = 0x0
time = 0x6a4c15b7 0xaba8869
eid = 0xd5
```

```
pool: rz3-2hs-16d
state: DEGRADED
status: One or more devices has been taken offline by the administrator.
Sufficient replicas exist for the pool to continue functioning in a degraded state.
action: Online the device using 'zpool online' or replace the device with 'zpool replace'.
scan: scrub repaired 0B in 00:00:15 with 0 errors
```

```
class=statechange pool='rz3-2hs-16d' vdev=
class=vdev_online pool='rz3-2hs-16d' vdev=
class=resilver_start pool='rz3-2hs-16d' vdev=
class=resilver_finish pool='rz3-2hs-16d' vdev=
```

# Disk Failure Simulation – Mock Environment

Degraded pool, induced more disk failures

```
pool: rz3-2hs-16d
state: DEGRADED
status: One or more devices is currently being resilvered. The pool will
        continue to function, possibly in a degraded state.
action: Wait for the resilver to complete.
scan: resilver in progress since Mon Jul 6 17:20:05 2026
      1.18G / 1.18G scanned, 763M / 1.18G issued at 44.9M/s
      42.4M resilvered, 63.11% done, 00:00:09 to go
```

| NAME               | STATE    | READ | WRITE | CKSUM |                 |
|--------------------|----------|------|-------|-------|-----------------|
| rz3-2hs-16d        | DEGRADED | 0    | 0     | 0     |                 |
| raidz3-0           | DEGRADED | 0    | 0     | 0     |                 |
| zvol/zvolumes/v048 | OFFLINE  | 0    | 0     | 0     |                 |
| zvol/zvolumes/v049 | OFFLINE  | 0    | 0     | 0     |                 |
| zvol/zvolumes/v050 | OFFLINE  | 0    | 0     | 0     |                 |
| spare-3            | DEGRADED | 0    | 0     | 0     |                 |
| zvol/zvolumes/v051 | DEGRADED | 0    | 0     | 0     | too many errors |
| zvol/zvolumes/v062 | ONLINE   | 0    | 0     | 0     |                 |
| spare-4            | DEGRADED | 0    | 0     | 0     |                 |
| zvol/zvolumes/v052 | DEGRADED | 0    | 0     | 0     | too many errors |
| zvol/zvolumes/v063 | ONLINE   | 0    | 0     | 0     | (resilvering)   |

|                    |       |                  |
|--------------------|-------|------------------|
| spares             |       |                  |
| zvol/zvolumes/v062 | INUSE | currently in use |
| zvol/zvolumes/v063 | INUSE | currently in use |

Note the 2  
spares  
kicking in  
for disks  
51 and 52



# Disk Failure Simulation – Mock Environment

Degraded pool, induced more disk failures

6 disk failures, no sufficient replicas to save the pool, even with the 2 hot spares in use

## DATA LOSS

This illustrative permutation for data loss shows 6 disks...

Stick to the conservative 3 parity with a 4<sup>th</sup> failed = data loss, the hot spare only provides borrowed time\*

```
pool: rz3-2hs-16d
state: DEGRADED
status: One or more devices has experienced an unrecoverable error. An
attempt was made to correct the error. Applications are unaffected.
action: Determine if the device needs to be replaced, and clear the errors
using 'zpool clear' or replace the device with 'zpool replace'.
see: https://openzfs.github.io/openzfs-docs/msg/ZFS-8000-9P
scan: resilvered 87.6M in 00:01:02 with 0 errors on Mon Jul  6 17:21:07 2026
config:

    NAME                                STATE        READ  WRITE  CKSUM
    rz3-2hs-16d                         DEGRADED      0      0      0
    raidz3-0                             DEGRADED      0      0      0
    zvol/zvolumes/v048                  OFFLINE       0      0      0
    zvol/zvolumes/v049                  OFFLINE       0      0      0
    zvol/zvolumes/v050                  OFFLINE       0      0      0
    spare-3                             DEGRADED      0      0      0
    zvol/zvolumes/v051                  DEGRADED      0      0      0  too many errors
    zvol/zvolumes/v062                  ONLINE        0      0      0
    spare-4                             DEGRADED      0      0      0
    zvol/zvolumes/v052                  DEGRADED      0      0      0  too many errors
    zvol/zvolumes/v063                  ONLINE        0      0      0
    zvol/zvolumes/v053                  DEGRADED      0      0      0  too many errors
    zvol/zvolumes/v054                  ONLINE        0      0      0
    zvol/zvolumes/v055                  ONLINE        0      0      0
    zvol/zvolumes/v056                  ONLINE        0      0      0
    zvol/zvolumes/v057                  ONLINE        0      0      0
    zvol/zvolumes/v058                  ONLINE        0      0      0
    zvol/zvolumes/v059                  ONLINE        0      0      0
    zvol/zvolumes/v060                  ONLINE        0      0      0
    zvol/zvolumes/v061                  ONLINE        0      0      0
    spares
    zvol/zvolumes/v062                  INUSE         currently in use
    zvol/zvolumes/v063                  INUSE         currently in use

errors: No known data errors
```



# Disk Failure Simulation

## – Mock Environment

Note the spares kicking in with the disks fault, in addition to the rebuild from the 1 or 2 disk worth of capacity, the other additional volumes (resilvering) are actively rebuilding against the raid pool

```
pool: drz3-2hs-16d
state: ONLINE
scan: resilvered 15.0M in 00:00:11 with 0 errors on Mon Jul 6 16:02:36 2026
config:

NAME                STATE READ WRITE CKSUM
drz3-2hs-16d         ONLINE  0     0     0
draid3:11d:16c:2s-0  ONLINE  0     0     0
zvol/zvolumes/v064   ONLINE  0     0     0
zvol/zvolumes/v065   ONLINE  0     0     0
zvol/zvolumes/v066   ONLINE  0     0     0
zvol/zvolumes/v067   ONLINE  0     0     0
zvol/zvolumes/v068   ONLINE  0     0     0
zvol/zvolumes/v069   ONLINE  0     0     0
zvol/zvolumes/v070   ONLINE  0     0     0
zvol/zvolumes/v071   ONLINE  0     0     0
zvol/zvolumes/v072   ONLINE  0     0     0
zvol/zvolumes/v073   ONLINE  0     0     0
zvol/zvolumes/v074   ONLINE  0     0     0
zvol/zvolumes/v075   ONLINE  0     0     0
zvol/zvolumes/v076   ONLINE  0     0     0
zvol/zvolumes/v077   ONLINE  0     0     0
zvol/zvolumes/v078   ONLINE  0     0     0
zvol/zvolumes/v079   ONLINE  0     0     0

spares
draid3-0-0           AVAIL
draid3-0-1           AVAIL

errors: No known data errors
```

```
NAME                STATE READ WRITE CKSUM
drz3-2hs-16d         DEGRADED  0     0     0
draid3:11d:16c:2s-0  DEGRADED  0     0     0
spare-0              DEGRADED  0     0     0
zvol/zvolumes/v064   FAULTED   0     0     0 too many errors
draid3-0-0           ONLINE    0     0     0
spare-1              DEGRADED  0     0     0
zvol/zvolumes/v065   FAULTED   0     0     0 too many errors
draid3-0-1           ONLINE    0     0     0 (resilvering)
zvol/zvolumes/v066   ONLINE    0     0     0 (resilvering)
zvol/zvolumes/v067   ONLINE    0     0     0 (resilvering)
```

```
pool: drz3-2hs-16d
state: DEGRADED
status: One or more devices is currently being resilvered. The pool will
continue to function, possibly in a degraded state.
action: Wait for the resilver to complete.
scan: scrub repaired 0B in 00:00:24 with 0 errors on Mon Jul 6 17:50:32 2026
scan: resilver (draid3:11d:16c:2s-0) in progress since Mon Jul 6 17:50:35 2026
413M / 737M scanned at 5.85M/s, 413M / 737M issued at 5.85M/s
16.1M resilvered, 56.08% done, no estimated completion time
```

|                    |        |   |   |   |               |
|--------------------|--------|---|---|---|---------------|
| zvol/zvolumes/v075 | ONLINE | 0 | 0 | 0 | (resilvering) |
| zvol/zvolumes/v076 | ONLINE | 0 | 0 | 0 | (resilvering) |
| zvol/zvolumes/v077 | ONLINE | 0 | 0 | 0 | (resilvering) |
| zvol/zvolumes/v078 | ONLINE | 0 | 0 | 0 | (resilvering) |
| draid3-0-1         | ONLINE | 0 | 0 | 0 | (resilvering) |
| zvol/zvolumes/v066 | ONLINE | 0 | 0 | 0 | (resilvering) |
| zvol/zvolumes/v067 | ONLINE | 0 | 0 | 0 | (resilvering) |

```
spares
draid3-0-0           INUSE      currently in use
draid3-0-1           INUSE      currently in use

errors: No known data errors
```

# Disk Failure Simulation – Mock Environment

Different permutations of the mock environment data destruction illustration

This one happens to be showing the rest of the volumes in the pool contributing to the rebuilding of the data pool integrity

```
pool: drz3-2hs-16d
state: ONLINE
status: One or more devices is currently being resilvered. The pool will
        continue to function, possibly in a degraded state.
action: Wait for the resilver to complete.
scan: resilver in progress since Mon Jul  6 18:06:53 2026
      728M / 738M scanned, 725M / 735M issued at 55.7M/s
      205M resilvered, 98.66% done, 00:00:00 to go

config:

      NAME                                STATE      READ  WRITE CKSUM
drz3-2hs-16d                               ONLINE         0     0     0
      draid3:11d:16c:2s:16c:16c:16c:16c ONLINE         0     0     0
      zvol/zvolumes/v064                   ONLINE         0     0     0      (resilvering)
      spare-1                               ONLINE         0     0     0
      zvol/zvolumes/v065                     ONLINE         0     0     0      (resilvering)
      draid3-0-1                             ONLINE         0     0     0      (resilvering)
      zvol/zvolumes/v066                     ONLINE         0     0     0      (resilvering)
      zvol/zvolumes/v067                     ONLINE         0     0     0      (resilvering)
      zvol/zvolumes/v068                     ONLINE         0     0     0      (resilvering)
      zvol/zvolumes/v069                     ONLINE         0     0     0      (resilvering)
      zvol/zvolumes/v070                     ONLINE         0     0     0      (resilvering)
      zvol/zvolumes/v071                     ONLINE         0     0     0      (resilvering)
      zvol/zvolumes/v072                     ONLINE         0     0     0      (resilvering)
      zvol/zvolumes/v073                     ONLINE         0     0     0      (resilvering)
      zvol/zvolumes/v074                     ONLINE         0     0     0      (resilvering)
      zvol/zvolumes/v075                     ONLINE         0     0     0      (resilvering)
      zvol/zvolumes/v076                     ONLINE         0     0     0      (resilvering)
      zvol/zvolumes/v077                     ONLINE         0     0     0      (resilvering)
      zvol/zvolumes/v078                     ONLINE         0     0     0      (resilvering)
      zvol/zvolumes/v079                     ONLINE         0     0     0      (resilvering)
      spares
      draid3-0-0                             AVAIL
      draid3-0-1                             INUSE      currently in use

errors: No known data errors
```



# Disk Failure Simulation – Mock Environment

Different permutations of the mock environment data destruction illustration

This one happens to be showing the data pool back to healthy and online

This may not always be the case – **ALWAYS BACK UP YOUR DATA**

```
pool: drz3-2hs-16d
state: ONLINE
scan: resilvered 268M in 00:00:37 with 0 errors on Mon Jul 6 18:07:30 2026
config:
```

| NAME                | STATE  | READ | WRITE | CKSUM |
|---------------------|--------|------|-------|-------|
| drz3-2hs-16d        | ONLINE | 0    | 0     | 0     |
| draid3:11d:16c:2s-0 | ONLINE | 0    | 0     | 0     |
| zvol/zvolumes/v064  | ONLINE | 0    | 0     | 0     |
| zvol/zvolumes/v065  | ONLINE | 0    | 0     | 0     |
| zvol/zvolumes/v066  | ONLINE | 0    | 0     | 0     |
| zvol/zvolumes/v067  | ONLINE | 0    | 0     | 0     |
| zvol/zvolumes/v068  | ONLINE | 0    | 0     | 0     |
| zvol/zvolumes/v069  | ONLINE | 0    | 0     | 0     |
| zvol/zvolumes/v070  | ONLINE | 0    | 0     | 0     |
| zvol/zvolumes/v071  | ONLINE | 0    | 0     | 0     |
| zvol/zvolumes/v072  | ONLINE | 0    | 0     | 0     |
| zvol/zvolumes/v073  | ONLINE | 0    | 0     | 0     |
| zvol/zvolumes/v074  | ONLINE | 0    | 0     | 0     |
| zvol/zvolumes/v075  | ONLINE | 0    | 0     | 0     |
| zvol/zvolumes/v076  | ONLINE | 0    | 0     | 0     |
| zvol/zvolumes/v077  | ONLINE | 0    | 0     | 0     |
| zvol/zvolumes/v078  | ONLINE | 0    | 0     | 0     |
| zvol/zvolumes/v079  | ONLINE | 0    | 0     | 0     |
| spares              |        |      |       |       |
| draid3-0-0          | AVAIL  |      |       |       |
| draid3-0-1          | AVAIL  |      |       |       |

# Disk Failure Simulation – Mock Environment

7 disk failures, no sufficient replicas to save the pool, even with the 2 hot spares in use

## DATA LOSS

This illustrative permutation for data loss shows 7 disks...

Stick to the conservative 3 parity with a 4<sup>th</sup> failed = data loss, the hot spare only provides borrowed time\*

```
pool: drz3-2hs-16d
state: DEGRADED
status: One or more devices has experienced an unrecoverable error. An
attempt was made to correct the error. Applications are unaffected.
action: Determine if the device needs to be replaced, and clear the errors
using 'zpool clear' or replace the device with 'zpool replace'.
see: https://openzfs.github.io/openzfs-docs/msg/ZFS-8000-9P
scan: resilver (draid3:11d:16c:2s-0) canceled on Mon Jul  6 18:23:00 2026
config:

NAME                                STATE      READ  WRITE CKSUM
drz3-2hs-16d                        DEGRADED   0      0     0
  draid3:11d:16c:2s-0               DEGRADED   0      0     0
    zvol/zvolumes/v064              OFFLINE    0      0     0
    zvol/zvolumes/v065              OFFLINE    0      0     0
    zvol/zvolumes/v066              OFFLINE    0      0     0
    spare-3                         DEGRADED   0      0     0
      zvol/zvolumes/v067             DEGRADED   0      0     0  too many errors
      draid3-0-0                    ONLINE     0      0     0
      zvol/zvolumes/v068             DEGRADED   0      0     0  too many errors
      spare-5                      DEGRADED   0      0     0
        zvol/zvolumes/v069          DEGRADED   0      0     0  too many errors
        draid3-0-1                 ONLINE     0      0     0
        zvol/zvolumes/v070          DEGRADED   0      0     0  too many errors
        zvol/zvolumes/v071          DEGRADED   0      0     0  too many errors
        zvol/zvolumes/v072          ONLINE     0      0     0
        zvol/zvolumes/v073          ONLINE     0      0     0
        zvol/zvolumes/v074          ONLINE     0      0     0
        zvol/zvolumes/v075          ONLINE     0      0     0
        zvol/zvolumes/v076          ONLINE     0      0     0
        zvol/zvolumes/v077          ONLINE     0      0     0
        zvol/zvolumes/v078          ONLINE     0      0     0
        zvol/zvolumes/v079          ONLINE     0      0     0
  spares
    draid3-0-0                      INUSE      currently in use
    draid3-0-1                      INUSE      currently in use

errors: No known data errors
```



# Disk Failure Simulation

## – Mock Environment

5 disk failures, no sufficient replicas to save the pool, even with the 2 hot spares in use

### DATA LOSS

This illustrative permutation for data loss shows 5 disks... more accurate...

Stick to the conservative 3 parity with a 4<sup>th</sup> failed = data loss, the hot spare only provides borrowed time\*

```
pool: drz3-2hs-24d
state: ONLINE
config:

NAME                                STATE      READ  WRITE CKSUM
drz3-2hs-24d
  draid3:19d:24c:2s-0              ONLINE      0     0     0
    zvol/zvolumes/v080             ONLINE      0     0     0
    zvol/zvolumes/v081             ONLINE      0     0     0
    zvol/zvolumes/v082             ONLINE      0     0     0
    zvol/zvolumes/v083             ONLINE      0     0     0
    zvol/zvolumes/v084             ONLINE      0     0     0
    zvol/zvolumes/v085             ONLINE      0     0     0
    zvol/zvolumes/v086             ONLINE      0     0     0
    zvol/zvolumes/v087             ONLINE      0     0     0
    zvol/zvolumes/v088             ONLINE      0     0     0
    zvol/zvolumes/v089             ONLINE      0     0     0
    zvol/zvolumes/v090             ONLINE      0     0     0
    zvol/zvolumes/v091             ONLINE      0     0     0
    zvol/zvolumes/v092             ONLINE      0     0     0
    zvol/zvolumes/v093             ONLINE      0     0     0
    zvol/zvolumes/v094             ONLINE      0     0     0
    zvol/zvolumes/v095             ONLINE      0     0     0
    zvol/zvolumes/v096             ONLINE      0     0     0
    zvol/zvolumes/v097             ONLINE      0     0     0
    zvol/zvolumes/v098             ONLINE      0     0     0
    zvol/zvolumes/v099             ONLINE      0     0     0
    zvol/zvolumes/v100             ONLINE      0     0     0
    zvol/zvolumes/v101             ONLINE      0     0     0
    zvol/zvolumes/v102             ONLINE      0     0     0
    zvol/zvolumes/v103             ONLINE      0     0     0
  spares
    draid3-0-0                     AVAIL
    draid3-0-1                     AVAIL

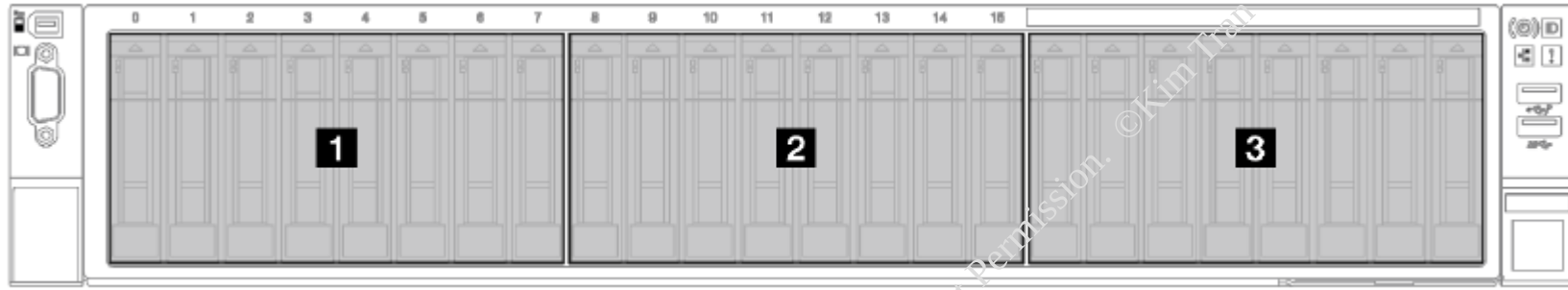
errors: No known data errors
```

```
pool: drz3-2hs-24d
state: DEGRADED
status: One or more devices has experienced an unrecoverable error. An
attempt was made to correct the error. Applications are unaffected.
action: Determine if the device needs to be replaced, and clear the errors
        using 'zpool clear' or replace the device with 'zpool replace'.
see: https://openzfs.github.io/openzfs-docs/msg/ZFS-8000-9P
scan: resilver (draid3:19d:24c:2s-0) canceled on Mon Jul  6 18:59:31 2026
config:

NAME                                STATE      READ  WRITE CKSUM
drz3-2hs-24d
  draid3:19d:24c:2s-0              DEGRADED      0     0     0
    zvol/zvolumes/v080             OFFLINE      0     0     0
    zvol/zvolumes/v081             OFFLINE      0     0     0
    zvol/zvolumes/v082             OFFLINE      0     0     0
  spare-3
    zvol/zvolumes/v083             DEGRADED      0     0     0  too many errors
    draid3-0-0                     ONLINE      0     0     0
    zvol/zvolumes/v084             DEGRADED      0     0     0  too many errors
```

# Storage Architecture Redesign based off of the existing equipment: SR850 V3

Machine Type: 7D96, 7D97, and 7D98



The server supports up to three 2.5-inch drive backplanes with the following corresponding drive backplane numbers: 1, 2, 3

When one or more of the following components are installed in the system, the maximum number of supported drive backplanes is two.

- ThinkSystem Mellanox ConnectX-6 Dx 100GbE QSFP56 2-port PCIe Ethernet Adapter
- ThinkSystem Nvidia ConnectX-7 NDR200/HDR QSFP112 2-Port PCIe Gen5 x16 InfiniBand Adapter
- ThinkSystem NVIDIA ConnectX-7 NDR400 OSFP 1-Port PCIe Gen5 Adapter
- ThinkSystem 96GB TruDDR5 4800MHz (2Rx4) RDIMM
- ThinkSystem 128GB TruDDR5 4800MHz (4Rx4) 3DS RDIMM v1
- ThinkSystem 128GB TruDDR5 4800MHz (4Rx4) 3DS RDIMM v2

When one of the following components is installed in the system, the maximum number of supported drive backplanes is one.

- ThinkSystem 256GB TruDDR5 4800MHz (8Rx4) 3DS RDIMM v1
- ThinkSystem 256GB TruDDR5 4800MHz (8Rx4) 3DS RDIMM v2

## Approximate capacity against parity protection

**CURRENT EXISTING**  
RAID5 / md0 (8x Disks)  
15TB Enterprise Scale: 105 TB usable (120 TB raw)

### Minimum Recommended

RAID6 / RAID-Z2 (8x Disks)  
15TB Enterprise Scale: 90 TB usable (120 TB raw)

RAID6 / RAID-Z2 + 1 Hot Spare (8x Disks Total)  
15TB Enterprise Scale: 75 TB usable (120 TB raw)

RAID-Z3 (8x Disks)  
15TB Enterprise Scale: 75 TB usable (120 TB raw)

### Recommended

RAID-Z3 (16x Disks)  
15TB Enterprise Scale: 195 TB usable (240 TB raw)

RAID-Z3 + 2 Hot Spares (16x Disks Total)  
15TB Enterprise Scale: 165 TB usable (240 TB raw)

dRAID3 + 2 Distributed Hot Spares (16x Disks Total)  
15TB Enterprise Scale: 165 TB usable (240 TB raw)

dRAID3 + 2 Distributed Hot Spares (24x Disks Total)  
15TB Enterprise Scale: 285 TB usable (360 TB raw)